

Linear Transforms

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Defn. A **linear transform** T is a function from one vector space to another vector space that interacts nicely with addition and scalar multiplication.

That is, for all vectors u and v in the domain and all reals c :

- (1) $T(u + v) = T(u) + T(v)$, and
- (2) $T(cu) = cT(u)$.

Every Matrix Transform is a Linear Transform

Fact. *Every matrix transform is a linear transform.*

The Kernel of a Linear Transform

Defn. *The null space of a linear transform is the set of vectors that are mapped to 0; it is often called the **kernel**.*

For example, differentiation is a linear transform from the polynomial space $\mathbb{P}_n$ to the polynomial space $\mathbb{P}_{n-1}$. Its kernel is the set of all constants.

Facts about Linear Transforms

Fact. For any linear transform:

- 1) The kernel is a subspace of the domain.
- 2) The range is a subspace of the codomain.

Summary

A linear transform is a function from a vector space to a vector space that interacts nicely with addition and scalar multiplication. Every matrix transform is a linear transform.

The null space or kernel of a linear transform is the set of all vectors mapped to 0. The kernel is a subspace of the domain while the range is a subspace of the codomain.